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A APPENDIX

A.1 PAC BAYESIAN GENERALIZATION BOUND

Below, we state a generalization bound based on sharpness.

Theorem 2. *For any $\rho > 0$ and any distribution $\mathscr{D}$, with probability $1-\delta$ over the choice of the training set $\mathcal{S} \sim \mathscr{D}$,*

$$L_{\mathscr{D}}(\boldsymbol{w}) \leq \max_{\|\boldsymbol{\epsilon}\|_2 \leq \rho} L_{\mathcal{S}}(\boldsymbol{w}+\boldsymbol{\epsilon}) + \sqrt{\frac{k \log\left(1 + \frac{\|\boldsymbol{w}\|_2^2}{\rho^2}\left(1+\sqrt{\frac{\log(n)}{k}}\right)^2\right) + 4\log\frac{n}{\delta} + \tilde{O}(1)}{n-1}} \tag{4}$$

where $n = |\mathcal{S}|$, k is the number of parameters and we assumed $L_{\mathscr{D}}(\boldsymbol{w}) \leq \mathbb{E}_{\epsilon_i \sim \mathcal{N}(0,\rho)}[L_{\mathscr{D}}(\boldsymbol{w}+\boldsymbol{\epsilon})]$.

The condition $L_{\mathscr{D}}(\boldsymbol{w}) \leq \mathbb{E}_{\epsilon_i \sim \mathcal{N}(0,\rho)}[L_{\mathscr{D}}(\boldsymbol{w}+\boldsymbol{\epsilon})]$ means that adding Gaussian perturbation should not decrease the test error. This is expected to hold in practice for the final solution but does not necessarily hold for any $\boldsymbol{w}$.

Proof. First, note that the right hand side of the bound in the theorem statement is lower bounded by $\sqrt{k\log(1+\|\boldsymbol{w}\|_2^2/\rho^2)/(4n)}$ which is greater than 1 when $\|\boldsymbol{w}\|_2^2 > \rho^2(\exp(4n/k)-1)$. In that case, the right hand side becomes greater than 1 in which case the inequality holds trivially. Therefore, in the rest of the proof, we only consider the case when $\|\boldsymbol{w}\|_2^2 \leq \rho^2(\exp(4n/k)-1)$.

The proof technique we use here is inspired from Chatterji et al. (2020). Using PAC-Bayesian generalization bound McAllester (1999) and following Dziugaite & Roy (2017), the following generalization bound holds for any prior $\mathscr{P}$ over parameters with probability $1-\delta$ over the choice of the training set $\mathcal{S}$, for any posterior $\mathscr{Q}$ over parameters:

$$\mathbb{E}_{\boldsymbol{w}\sim\mathscr{Q}}[L_{\mathscr{D}}(\boldsymbol{w})] \leq \mathbb{E}_{\boldsymbol{w}\sim\mathscr{Q}}[L_{\mathcal{S}}(\boldsymbol{w})] + \sqrt{\frac{KL(\mathscr{Q}||\mathscr{P}) + \log\frac{n}{\delta}}{2(n-1)}} \tag{5}$$

Moreover, if $\mathscr{P} = \mathcal{N}(\boldsymbol{\mu}_P, \sigma_P^2\boldsymbol{I})$ and $\mathscr{Q} = \mathcal{N}(\boldsymbol{\mu}_Q, \sigma_Q^2\boldsymbol{I})$, then the KL divergence can be written as follows:

$$KL(\mathscr{P}||\mathscr{Q}) = \frac{1}{2}\left[\frac{k\sigma_Q^2 + \|\boldsymbol{\mu}_P - \boldsymbol{\mu}_Q\|_2^2}{\sigma_P^2} - k + k\log\left(\frac{\sigma_P^2}{\sigma_Q^2}\right)\right] \tag{6}$$

Given a posterior standard deviation σ_Q, one could choose a prior standard deviation σ_P to minimize the above KL divergence and hence the generalization bound by taking the derivative[10] of the above KL with respect to σ_P and setting it to zero. We would then have ${\sigma_P^*}^2 = \sigma_Q^2 + \|\boldsymbol{\mu}_P - \boldsymbol{\mu}_Q\|_2^2/k$. However, since σ_P should be chosen before observing the training data $\mathcal{S}$ and $\boldsymbol{\mu}_Q$,σ_Q could depend on $\mathcal{S}$, we are not allowed to optimize σ_P in this way. Instead, one can have a set of predefined values for σ_P and pick the best one in that set. See Langford & Caruana (2002) for the discussion around this technique. Given fixed $a, b > 0$, let $T = \{c\exp((1-j)/k) | j \in \mathbb{N}\}$ be that predefined set of values for σ_P^2. If for any $j \in \mathbb{N}$, the above PAC-Bayesian bound holds for $\sigma_P^2 = c\exp((1-j)/k)$ with probability $1-\delta_j$ with $\delta_j = \frac{6\delta}{\pi^2 j^2}$, then by the union bound, all above bounds hold simultaneously with probability at least $1 - \sum_{j=1}^{\infty}\frac{6\delta}{\pi^2 j^2} = 1-\delta$.

Let $\sigma_Q = \rho$, $\boldsymbol{\mu}_Q = \boldsymbol{w}$ and $\boldsymbol{\mu}_P = \boldsymbol{0}$. Therefore, we have:

$$\sigma_Q^2 + \|\boldsymbol{\mu}_P - \boldsymbol{\mu}_Q\|_2^2/k \leq \rho^2 + \|\boldsymbol{w}\|_2^2/k \leq \rho^2(1+\exp(4n/k)) \tag{7}$$

We now consider the bound that corresponds to $j = \lfloor 1 - k\log((\rho^2 + \|\boldsymbol{w}\|_2^2/k)/c)\rfloor$. We can ensure that $j \in \mathbb{N}$ using inequality equation 7 and by setting $c = \rho^2(1+\exp(4n/k))$. Furthermore, for $\sigma_P^2 = c\exp((1-j)/k)$, we have:

$$\rho^2 + \|\boldsymbol{w}\|_2^2/k \leq \sigma_P^2 \leq \exp(1/k)\left(\rho^2 + \|\boldsymbol{w}\|_2^2/k\right) \tag{8}$$

[10] Despite the nonconvexity of the function here in σ_P^2, it has a unique stationary point which happens to be its minimizer.

Therefore, using the above value for σ_P, KL divergence can be bounded as follows:

$$
\begin{aligned}
KL(\mathscr{P}||\mathscr{Q}) &= \frac{1}{2}\left[\frac{k\sigma_Q^2 + \|\boldsymbol{\mu}_P - \boldsymbol{\mu}_Q\|_2^2}{\sigma_P^2} - k + k\log\left(\frac{\sigma_P^2}{\sigma_Q^2}\right)\right] && (9)\\
&\leq \frac{1}{2}\left[\frac{k(\rho^2 + \|\boldsymbol{w}\|_2^2/k)}{\rho^2 + \|\boldsymbol{w}\|_2^2/k} - k + k\log\left(\frac{\exp(1/k)\left(\rho^2 + \|\boldsymbol{w}\|_2^2/k\right)}{\rho^2}\right)\right] && (10)\\
&= \frac{1}{2}\left[k\log\left(\frac{\exp(1/k)\left(\rho^2 + \|\boldsymbol{w}\|_2^2/k\right)}{\rho^2}\right)\right] && (11)\\
&= \frac{1}{2}\left[1 + k\log\left(1 + \frac{\|\boldsymbol{w}\|_2^2}{k\sigma_Q^2}\right)\right] && (12)
\end{aligned}
$$

Given the bound that corresponds to j holds with probability $1 - \delta_j$ for $\delta_j = \frac{6\delta}{\pi^2 j^2}$, the log term in the bound can be written as:

$$
\begin{aligned}
\log\frac{n}{\delta_j} &= \log\frac{n}{\delta} + \log\frac{\pi^2 j^2}{6}\\
&\leq \log\frac{n}{\delta} + \log\frac{\pi^2 k^2 \log^2(c/(\rho^2 + \|\boldsymbol{w}\|_2^2/k))}{6}\\
&\leq \log\frac{n}{\delta} + \log\frac{\pi^2 k^2 \log^2(c/\rho^2)}{6}\\
&\leq \log\frac{n}{\delta} + \log\frac{\pi^2 k^2 \log^2(1 + \exp(4n/k))}{6}\\
&\leq \log\frac{n}{\delta} + \log\frac{\pi^2 k^2 (2 + 4n/k)^2}{6}\\
&\leq \log\frac{n}{\delta} + 2\log(6n + 3k)
\end{aligned}
$$

Therefore, the generalization bound can be written as follows:

$$
\mathbb{E}_{\epsilon_i\sim\mathcal{N}(0,\sigma)}[L_{\mathscr{D}}(\boldsymbol{w}+\boldsymbol{\epsilon})] \leq \mathbb{E}_{\epsilon_i\sim\mathcal{N}(0,\sigma)}[L_{\mathcal{S}}(\boldsymbol{w}+\boldsymbol{\epsilon})] + \sqrt{\frac{\frac{1}{4}k\log\left(1 + \frac{\|\boldsymbol{w}\|_2^2}{k\sigma^2}\right) + \frac{1}{4} + \log\frac{n}{\delta} + 2\log(6n + 3k)}{n-1}} \tag{13}
$$

In the above bound, we have $\epsilon_i \sim \mathcal{N}(0, \sigma)$. Therefore, $\|\boldsymbol{\epsilon}\|_2^2$ has chi-square distribution and by Lemma 1 in Laurent & Massart (2000), we have that for any positive t:

$$
P(\|\boldsymbol{\epsilon}\|_2^2 - k\sigma^2 \geq 2\sigma^2\sqrt{kt} + 2t\sigma^2) \leq \exp(-t) \tag{14}
$$

Therefore, with probability $1 - 1/\sqrt{n}$ we have that:

$$
\|\boldsymbol{\epsilon}\|_2^2 \leq \sigma^2(2\ln(\sqrt{n}) + k + 2\sqrt{k\ln(\sqrt{n})}) \leq \sigma^2 k\left(1 + \sqrt{\frac{\ln(n)}{k}}\right)^2 \leq \rho^2
$$